

 Python Application Deployment on AWS EC2 using Apache & Gunicorn

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Summary

 Python Application Deployment on AWS EC2 using Apache & Gunicorn

Project Introduction

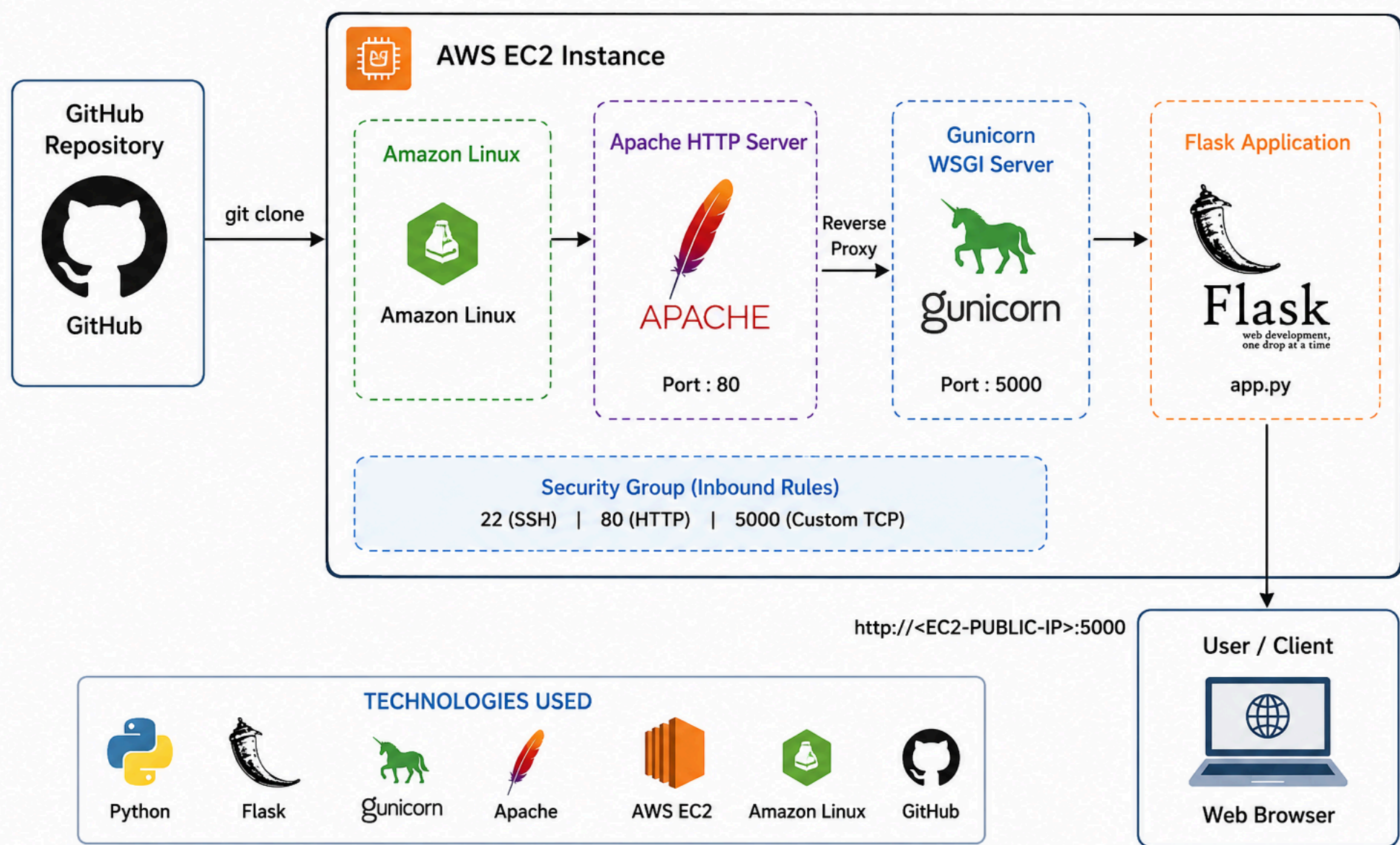
Python-App Deployment is a web-based application developed using Python, where the application runs using a built-in server and can also be deployed using Gunicorn for production-level performance.

Project objective:

Launching and configuring AWS EC2 instances
Hosting Python web applications on Linux servers
Managing Python virtual environments
Installing and configuring Apache HTTP Server
Running Flask applications using Gunicorn WSGI server
Configuring security groups and networking
Using GitHub for source code management

1. Architecture Diagram

ARCHITECTURE DIAGRAM



Technologies Used

- Python 3
- Flask
- Gunicorn
- AWS EC2
- Apache HTTP Server
- Amazon Linux
- Git & GitHub

Project Structure

```
pythonapp/  
├── app.py  
└── requirements.txt
```

2. Instance Setup

2.1 Launch Instance

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

Python

Add additional tags

▼ Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Q

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

Debian

>

Q

Browse more AMIs

Including AMIs from AWS Marketplace and the Community

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11...
ami-0236922087fa98b6e

Virtual server type (instance type)

t3.micro

Firewall (security group)

launch-wizard-2

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

CloudShell

Feedback

Console Mobile App

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2.2 Names & OS

Name : Python

OS : Amazon Linux

▼ Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

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Amazon Linux

aws

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Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

>

Q

Browse more AMIs

Including AMIs from AWS Marketplace and the Community

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

2.3 AMI & Instance Type

AMI : Default

Architecture : Default

Instance Type : t3.micro (2CPU & 1GB Memory)

EC2

>

Instances

>

Launch an instance

Amazon Machine Image (AMI)

Free tier eligible

Amazon Linux 2023 kernel-6.1 AMI

ami-098e39bafa7e7303d (64-bit (x86), uefi-preferred) / ami-03ed25db53d8de46c (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260413.0 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-098e39bafa7e7303d

Publish Date

2026-04-10

Username

ec2-user

Verified provider

▼ Instance type

Info | Get advice

Instance type

Free tier eligible

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11...
ami-098e39bafa7e7303d

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

04-22_180704.p...

2.4 Key & Network

Key : Pratik-Virginia-Key (Exiting One)

Security Group : Launch-Wizard-2 (Exiting One)

aws

Search

[Alt+S]

Ask Amazon Q

United States (N

EC2

Instances

Launch an instance

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Pratik-virginia-key

Create new key pair

▼ Network settings

Info

Edit

Network

Info

vpc-0c7849a9d6a173541

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.ami-0236922087fa98b6e

Virtual server type (instance t

t3.micro

Firewall (security group)

launch-wizard-2

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

2.5 Start Instance

aws

Search

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EC2

Instances

EC2

<

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

▼ Images

AMIs

AMI Catalog

Instances (1/8)

Info

Last updated less than a minute ago

Connect

Instance state

Action

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status c
☒	Python	i-07ea4a2051a261d7b	Stopped	t3.micro	-
☐	Virgina	i-0753e3c57c5933198	Stopped	t3.micro	-
☐	Registration-F...	i-0f9a75c1638b7f859	Stopped	t3.micro	-
☐	node-server	i-0f3f9ec5650065a48	Stopped	t3.micro	-

i-07ea4a2051a261d7b (Python)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary

Info

Instance ID

i-07ea4a2051a261d7b

Public IPv4 address

-

Private IPv4 addresses

172.31.31.7

IPv6 address

Instance state

Public DNS

2.6 Connect

aws

Search

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Ask Amazon Q

United States (N

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Instances (1/8)

Info

Last updated less than a minute ago

Connect

Instance state

Action

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate (delete) instance

us-east-1d

us-east-1d

	Name	Instance ID	Instance state	Instance type	Status c
☒	Python	i-07ea4a2051a261d7b	Stopped	t3.micro	-
☐	Virgina	i-0753e3c57c5933198	Stopped	t3.micro	-
☐	Registration-F...	i-0f9a75c1638b7f859	Stopped	t3.micro	-
☐	node-server	i-0f3f9ec5650065a48	Stopped	t3.micro	-

i-07ea4a2051a261d7b (Python)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-07ea4a2051a261d7b

IPv6 address

Public IPv4 address

-

Instance state

Stopped

Private IPv4 addresses

172.31.31.7

Public DNS

2.7 Copy SSH Client Link

EC2 > Instances > i-07ea4a2051a261d7b > Connect to instance

Instance ID

i-07ea4a2051a261d7b (Python)

VPC ID

vpc-0c7849a9d6a173541

Security groups

sg-018b1d7b7d7dae886 (launch-wizard-2)

IAM role

-

EC2 Instance Connect

SSM Session Manager

SSH client

EC2 serial console

Instance ID

i-07ea4a2051a261d7b (Python)

1. Open an SSH client.

2. Locate your private key file. The key used to launch this instance is Pratik-virginia-key.pem

3. Run this command, if necessary, to ensure your key is not publicly viewable.

chmod 400 "Pratik-virginia-key.pem"

4. Connect to your instance using its Public DNS:

ec2-18-205-237-57.compute-1.amazonaws.com

Command copied

ssh -i "Pratik-virginia-key.pem" ec2-user@ec2-18-205-237-57.compute-1.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

3. Installing Packages

3.1 Connect Instance With Public IP

```
ssh -i Pratik-Virginia-Key.pem ec2-user@54.161.38.170
```

3.2 Update Packages

```
sudo yum update
```

3.3 Install Git

```
sudo yum install git -y
```

```
[ec2-user@Python ~]$ pwd
/home/ec2-user
[ec2-user@Python ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository                289 kB/s | 3
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@Python ~]$ sudo yum install git -y
Last metadata expiration check: 0:00:34 ago on Mon May 18 12:47:45 2026.
Dependencies resolved.

=====
Package                                Architecture    Version                                Repository
=====
Installing:
git                                    x86_64          2.50.1-1.amzn2023.0.1                 amazonli
Installing dependencies:
git-core                             x86_64          2.50.1-1.amzn2023.0.1                 amazonli
git-core-doc                         noarch          2.50.1-1.amzn2023.0.1                 amazonli
perl-Error                           noarch          1:0.17030-2.amzn2023.0.1             amazonli
perl-File-Find                      noarch          1.37-477.amzn2023.0.8                amazonli
perl-Git                             noarch          2.50.1-1.amzn2023.0.1                 amazonli
perl-TermReadKey                    x86_64          2.38-9.amzn2023.0.2                  amazonli
perl-lib                             x86_64          0.65-477.amzn2023.0.8                amazonli

Transaction Summary
=====
Install 8 Packages

Total download size: 7.9 M
Installed size: 41 M
```

3.4 Git Clone Repo

```
git clone https://github.com/iamtruptimane/pythonapp
```

3.5 Install Python

```
sudo yum install python3 -y
```

```
[ec2-user@Python pythonapp]$ sudo yum install python3 -y
Last metadata expiration check: 0:13:29 ago on Mon May 18 12:47:45 2026.
Package python3-3.9.25-1.amzn2023.0.5.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

3.6 Install PIP

```
sudo yum install python3-pip -y
```

3.7 View Git, Python & PIP Version

```
git --versio
python --version
pip --version
```

```
[ec2-user@Python ~]$ git --version
git version 2.50.1
```

4. Running Python App

4.1 Go To Directory Which Create Via Git Clone

```
cd / pythonapp;
```

```
[ec2-user@python pythonapp]$ ls
app.py  requirements.txt  test
```

4.2 Create Virtual Environment (venv) Inside Project Folder

```
sudo python3 -m venv myenv
```

```
[ec2-user@Python pythonapp]$ python3 -m venv myenv
[ec2-user@Python pythonapp]$ source myenv/bin/activate
(myenv) [ec2-user@Python pythonapp]$ |
```

4.3 Bash Activated venv

```
sudo bash myenv/bin/activate
```

4.4 Install Requirements Inside Project Folder

```
sudo pip install -r requirements.txt
```

4.5 To Run App (FrontGround)

```
python3 app.py

If Press Ctrl + C → App Stops
```

4.6 To Run App In BackGround

```
gunicorn --bind 0.0.0.0:5000 app:app --daemon
```

5. Access On Browser

5.1 To Add Port Number 5000

5.1.1 Go To Security Groups

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

sg-018b1d7b7d7dae886 - launch-wizard-2

Details

Security group name

launch-wizard-2

Security group ID

sg-018b1d7b7d7dae886

Description

launch-wizard-2 created 2026-02-11T10:51:41.707Z

Owner

866059265852

Inbound rules count

5 Permission entries

Outbound rules count

1 Permission entry

VPC ID

vpc-0

5.1.2 Edit Inbound Rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - opti
sgr-089337022316ffc75	Custom TCP	TCP	85	Custom	0.0.0.0/0
sgr-0a221a12d679d1024	Custom TCP	TCP	5000	Custom	0.0.0.0/0
sgr-028fff770c70789cd	Custom TCP	TCP	3000	Custom	0.0.0.0/0
sgr-07fba8c57fe04a17a	SSH	TCP	22	Custom	0.0.0.0/0
sgr-0c42fbceb0218b58f	HTTP	TCP	80	Custom	0.0.0.0/0

5.1.3 Add Rule → Type : Custom TCP → Port Range : 3000 → Source : Anywhere-IPv4 → Save Rules

aws

Search

[Alt+S] Ask Amazon Q

United States (N

☰

EC2 > Security Groups > sg-018b1d7b7d7dae886 - launch-wizard-2 > Edit inbound rules

Edit inbound rules

Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Info

Security group rule ID	Type	Info	Protocol	Info	Port range	Source	Info	Description - opt
sgr-0a221a12d679d1024	Custom TCP	▼	TCP		5000	Custom	▼	0.0.0.0/0 ×
sgr-07fba8c57fe04a17a	SSH	▼	TCP		22	Custom	▼	0.0.0.0/0 ×
sgr-0c42fbceb0218b58f	HTTP	▼	TCP		80	Custom	▼	0.0.0.0/0 ×

Add rule

⚠️

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

CloudShell

Feedback

Console Mobile App

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5.2 Hit It On Browser And Put File Name

public ip:5000

54.91.157:5000

×

+

←

→

↺

⚠️ Not secure

54.91.150.157:5000

successfully deployed python application through jenkins!!!!!!!!!!, added webhook

- # 🔥 Key Learnings
- AWS EC2 setup

- Linux server management

- Python virtual environments

- Flask application deployment

- Gunicorn configuration

- Apache server basics

- GitHub integration

Summary

It Enables Users To Access The Application Through A Public IP On Port 5000, Demonstrating Basic Cloud Deployment And Application Hosting In A Structured Environment.